AWS

HEROVIRED · INFRASTRUCTURE AS CODE ASSIGNMENT

EC2 Nginx Deployment with Terraform

A minimal, secure and reproducible Ubuntu 20.04 Nginx deployment in the existing AWS default VPC.

SUBMITTED BY
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CLOUD PLATFORM
AWS · ap-south-1

IAC TOOL
Terraform 1.15.8

COMPUTE
Ubuntu 20.04 LTS · t2.micro

REPOSITORY
TabrezAjaz/Terraform-Nginx-EC2-Ubuntu

TERRAFORM · AWS EC2 · UBUNTU · NGINX · DEFAULT VPC

Abstract

This report documents the complete implementation of the assignment *EC2 Nginx Deployment with Terraform*. Terraform was used to query the existing AWS default VPC and default subnet, create one security group, and launch one official Canonical Ubuntu 20.04 LTS `t2.micro` instance. EC2 `user_data` installed Nginx and replaced its default page with the exact required welcome message.

The server was deployed live, its public IP was produced through Terraform output, and the custom page was verified over HTTP. Genuine evidence was captured before `terraform destroy` removed both managed resources. The final Terraform state is empty and the EC2 instance is confirmed terminated.

2	HTTP 200	2 destroyed
Resources created	Custom page verified	Zero resources active

> **Result:** every assignment requirement and marking criterion was implemented and verified.

1. Requirement Traceability

Requirement	Implementation	Status
AWS provider and region	AWS provider 5.x with configurable `aws_region` and profile	**Complete**
Ubuntu 20.04 LTS	Latest official Canonical Focal AMI data lookup	**Verified**
t2.micro in default VPC	Exact type enforced by variable validation; existing VPC/subnet data sources	**Verified**
HTTP 80 and SSH 22	Public HTTP; SSH restricted to administrator `/32`	**Complete**
Install Nginx using user_data	APT update/install, custom HTML write, enable and restart service	**Verified**
Required page content	Exact message returned from the public EC2 address	**Verified**
Output public IP	`public_ip` and `nginx_url` outputs	**Complete**
Proper teardown	Terraform destroyed EC2 and security group; state empty	**Verified**
README documentation	Run steps, resources, security, evidence and teardown included	**Complete**

2. Architecture and Design



The implementation deliberately reuses the account's existing default VPC and default subnet. Terraform creates no VPC, subnet, route table, Internet Gateway or NAT Gateway. This keeps the solution aligned with the task, simple to review and inexpensive to execute.

Managed Resources

Resource	Purpose	Key Configuration
`aws_security_group.nginx`	Network boundary	HTTP 80 public; SSH 22 from administrator CIDR; outbound package access
`aws_instance.nginx`	Nginx host	Ubuntu 20.04, t2.micro, public IPv4, encrypted 8 GiB gp3, IMDSv2

Existing Resources Queried

`aws_vpc.default`, `aws_subnet.default_1a`, and `aws_ami.ubuntu_2004` are data sources only and are never created or destroyed.

3. Terraform Implementation

AMI Selection

The configuration requests the latest x86_64 official Canonical image matching `ubuntu-focal-20.04-amd64-server-*`. The verified deployment resolved AMI `ami-06cc5ebfb8571a147`.

Security Group

HTTP is publicly available as required. SSH is safer than the minimum requirement because it is restricted to the administrator's current public address instead of exposing port 22 globally.

User Data

```
#!/bin/bash
set -e
apt-get update
apt-get install -y nginx
cat > /var/www/html/index.html <<'HTML'
...
Welcome to the Terraform-managed Nginx Server on Ubuntu
...
HTML
systemctl enable nginx
systemctl restart nginx
```

Outputs

The configuration exposes `public_ip`, `nginx_url`, `instance_id`, and `default_vpc_id` after apply.

4. Terraform CLI Execution

Only the required Terraform CLI workflow was used:

```
terraform init
terraform plan -out=assignment.tfplan
terraform apply assignment.tfplan
terraform output -raw public_ip
terraform destroy -auto-approve
```

The reviewed plan created only the EC2 instance and security group. The initial default-subnet selection resolved to `ap-south-1c`, where legacy `t2.micro` capacity is unsupported. The configuration was corrected to select the existing default subnet in `ap-south-1a`; no subnet was created.

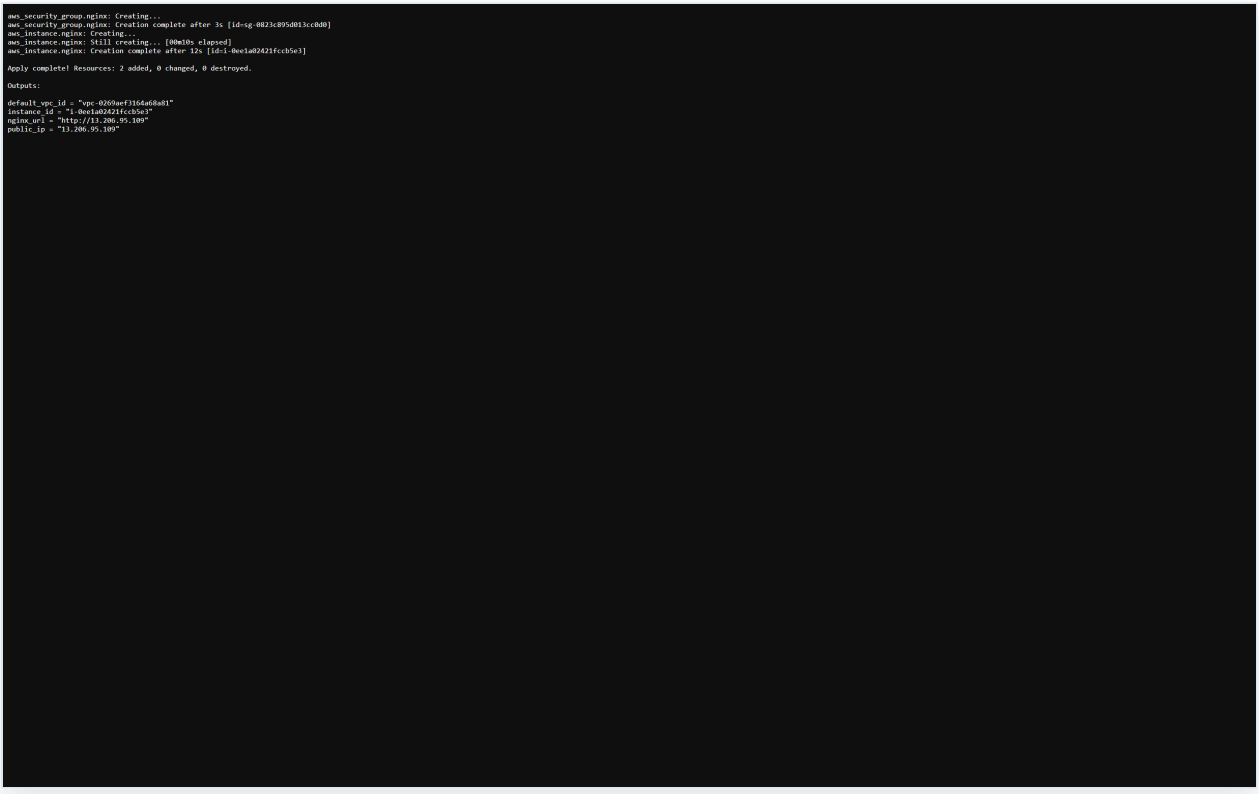


Figure 4.1 — Terraform apply outputs and verified AWS resource identifiers.

5. Public Nginx Verification

The public URL `http://13.206.95.109` returned HTTP 200 and the exact required custom content. The visual design below was written entirely by EC2 user data to `/var/www/html/index.html`.

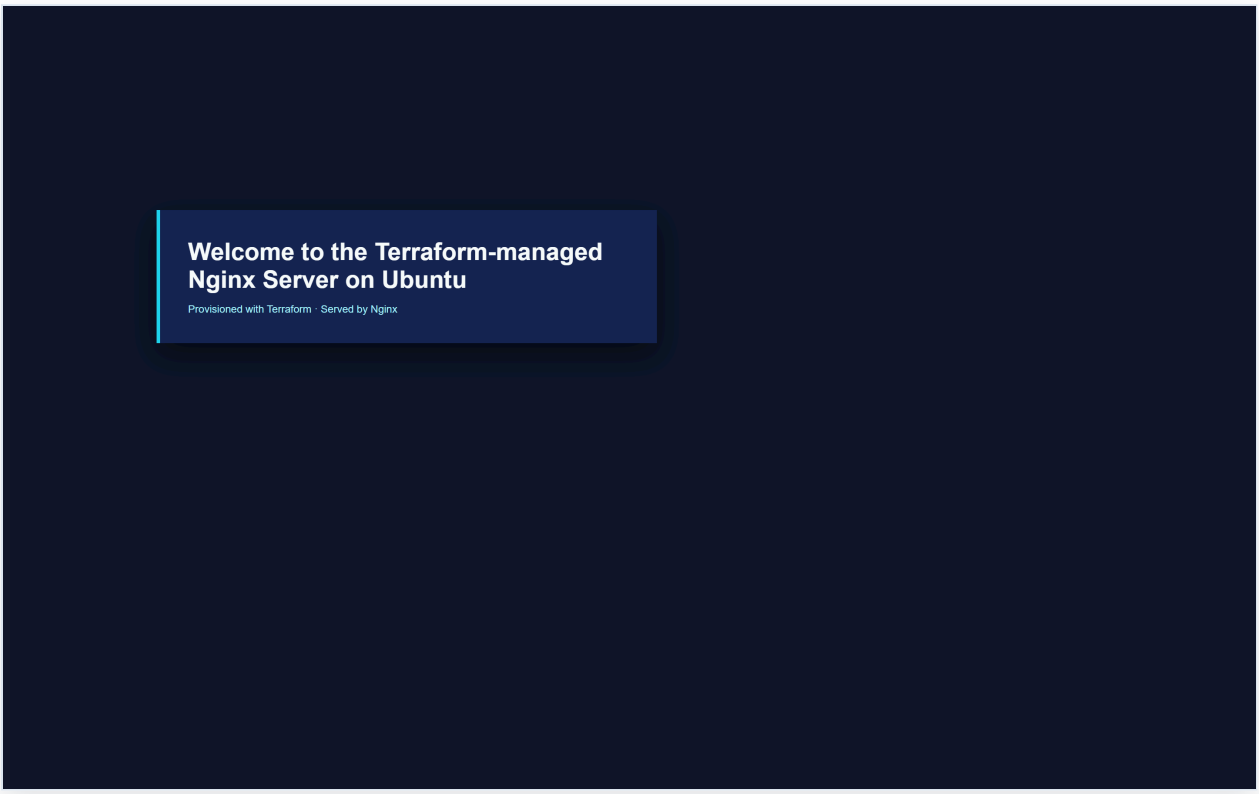


Figure 5.1 — Authentic browser capture of the required Nginx welcome page on the Terraform-managed Ubuntu instance.

6. Security and Best Practices

- All resources carry Project, ManagedBy, Owner and Name tags.
- SSH is restricted to a single administrator CIDR.
- EC2 Instance Metadata Service requires IMDSv2 tokens.
- The encrypted gp3 root volume is deleted with the instance.
- Terraform state, saved plans, real variables, PEM files and crash logs are Git-ignored.
- `create_before_destroy` avoids security-group naming conflicts during replacement.
- Variable validation prevents accidental use of any instance type other than the required `t2.micro`.

> **Cost control:** the EC2 instance was destroyed immediately after evidence capture.

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7. Teardown Verification

`terraform destroy` removed the instance, its encrypted root volume and the assignment security group. The existing default VPC and subnet were data sources and therefore remained untouched.



Figure 7.1 — Authentic Terraform CLI output showing successful cost-safe teardown.

Check	Observed Result
Terraform state	**Empty**
EC2 instance `i-0ee1a02421fccb5e3`	**terminated**
Destroy summary	**Destroy complete! Resources: 2 destroyed.**

8. Repository and Submission

The public repository contains the Terraform source, provider lock file, example variables, README, original assignment paper and genuine evidence screenshots.

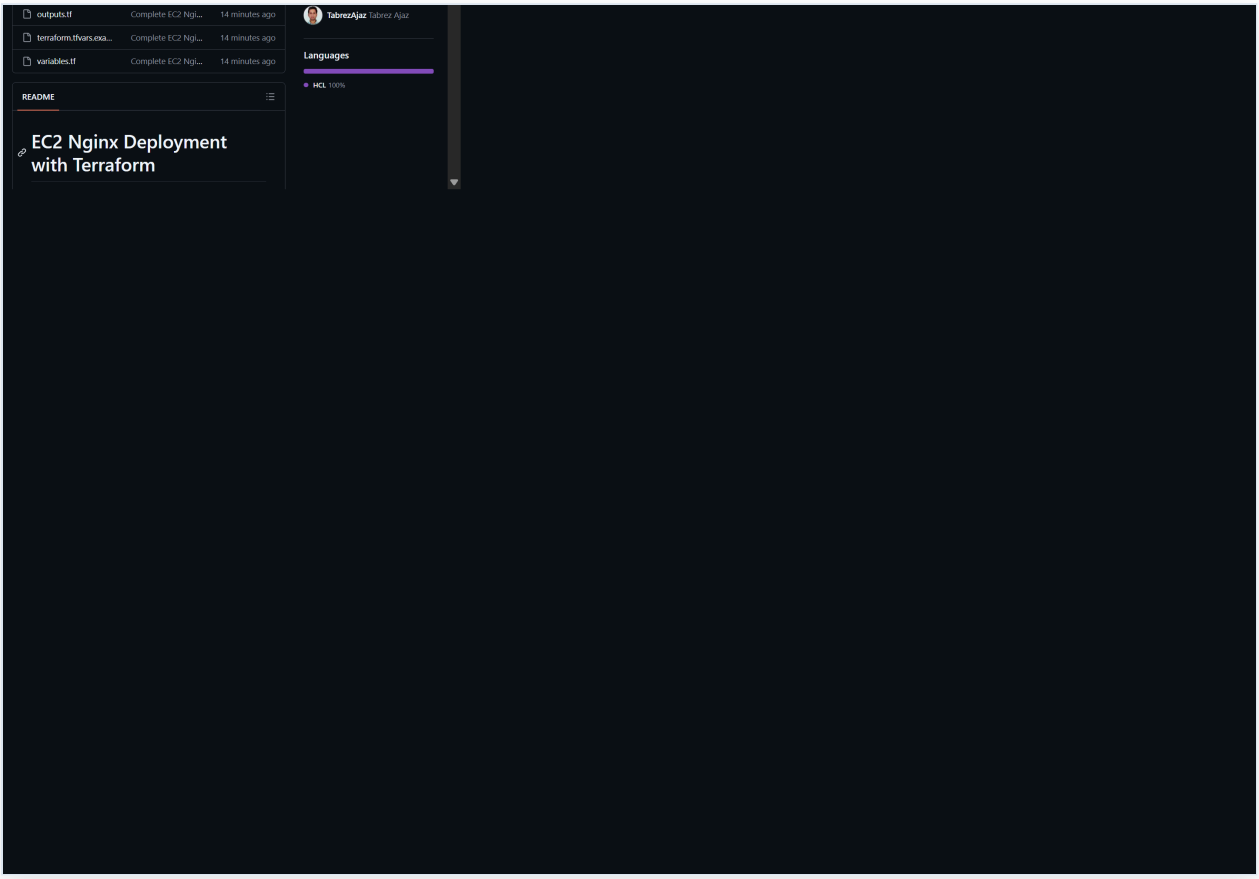


Figure 8.1 — Public GitHub submission repository.

Deliverable	Location
GitHub repository	github.com/TabrezAjaz/Terraform-Nginx-EC2-Ubuntu
Terraform configuration	`main.tf`, `variables.tf`, `outputs.tf`
Execution documentation	`README.md`
Evidence	`docs/screenshots/`
Portal link file	`SUBMISSION_LINK.txt`

> **Submission status: COMPLETE.** All 50-mark criteria are covered with implementation

> and verified evidence.

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9. Conclusion

The assignment was completed with a minimal Terraform design that creates only the two required managed resources. An official Ubuntu 20.04 `t2.micro` instance was launched in the existing default VPC, Nginx was installed and customized automatically, and the required page was verified through the Terraform public-IP output. Finally, Terraform destroyed all created resources and left an empty state.